Alejandro Enriquez

Embedded Software Engineer

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Summary

Embedded Software Engineer with 4 years developing safety-critical firmware on CAN-based multi-ECU architectures. Proven ability to own problems end-to-end, from hardware bring-up to manufacturing support. Delivered a cross-generation Software Thread Watchdog, reduced build times by ~65%, and unblocked stalled production lines. Experienced in V-model, HIL, and cross-site collaboration.

Key Achievements

- 65% faster builds (20+ min to 7 min)
- 2 production-critical products unblocked
- 3+ years shipping safety-critical firmware

Professional Experience

Firmware & Embedded Software Engineer — *Carrier Transicold | Monterrey Technology Hub | May 2025 – Present*

- Established the site's firmware development capability from zero. First Transicold embedded software engineer at the Monterrey Technology Hub.

- Developed firmware following a V-model process on a safety-critical multi-ECU refrigeration platform, covering implementation and validation across ARM Cortex-M4 (embOS) and ARM Cortex-M7 (MQX) targets.

- Designed and implemented a Software Thread Watchdog (STW) for two hardware generations (ARM Cortex-M4/embOS, ARM Cortex-M7/MQX), iterating from a timestamp-based design to a counter-based approach after identifying stack issues and false alarms caused by critical sections. Features: fault injection API, auto-registering tasks, per-task counters, and CAN-bus diagnostic messages with task ID, ASCII identifier, missed-count, and a custom marker.

- Traveled to Syracuse, NY to train on testbench setup and perform initial STW hardware validation, then brought back CAN bus sniffers, debuggers, and electronic boards to equip the Monterrey site.

- Built the site's first physical testbench from scratch, validating 5+ ECUs across 2 product generations.

- Unblocked two truck/trailer refrigeration products stalled for months in PPAP: one caused by a missing CAN activation sequence in manufacturing tests; the other by incorrect firmware being flashed due to board version confusion and an outdated Windchill PLM configuration.

- Developed a Python build automation script replacing a multi-IDE, multi-step manual compilation process, reducing build time from 20+ minutes to ~7 minutes. In active use by the team (v1.1 in progress).

- Performed low-level debugging and validation using J-Link, Kvaser, and Peak, including CAN/J1939 bus analysis and real-time diagnostics.

Embedded Software Engineer — *Terex Corporation | Monterrey, Nuevo León | Mar 2023 – May 2025*

- Developed firmware for a new rough terrain scissor lift prototype (Mule) from scratch. Implemented platform lift, lower, and outrigger actuation on a CAN-based multi-ECU architecture using IBM Rhapsody UML statecharts.

- Proposed and implemented a redesigned hydraulic software architecture, decomposing a monolithic handler into a dedicated statechart class with event-driven coupling, improving modularity and reducing inter-component dependencies.

- Developed the Kubota engine driver (ECON module) and built a CAN node simulator in BUSMASTER to keep development moving while a faulty magnetic relay in the Monterrey unit was being diagnosed.

- Implemented per-wheel PID torque balancing for the hybrid drivetrain.
- Recovered a prototype that arrived non-functional in a shipping container under deadline pressure, diagnosing cascading issues (discharged battery, multiple CAN errors, erratic behavior) and tracing the root cause to a phantom load cell module incorrectly connected during factory build.

- Authored Polarion requirements for the full scissor lift family and linked them to test cases; led 30+ test case executions on physical machines, logging failures and driving fixes through full fix-verify cycles.

- Performed HIL testing using a full machine simulator, validating firmware across multiple scissor lift variants, catching integration bugs before factory deployment.

- Led cross-site knowledge transfer between Monterrey, Seattle HQ, and the Changzhou factory, collaborating with core engineering and manufacturing teams across three countries.

Embedded Software Engineer (Intern) — *John Deere | Monterrey, Nuevo León | Jul 2022 – Mar 2023*

- Supported firmware development for a CAN + BLE master-slave module pair on a forestry machine, enabling bidirectional communication between a grapple arm and the driver cabin; debugged hardware using UART, JTAG, and direct MCU interfaces.
- Proposed and implemented a PowerShell build script improvement combining two separate firmware builds into a single step using preprocessor directives, reducing build friction for the team.

Education

B.S. in Digital Systems and Robotics Engineering | *Tecnológico de Monterrey (ITESM) | Nuevo León, México* | *2023*

Full Scholarship "Líderes del Mañana" – awarded for academic excellence and leadership potential.

Technical Competencies

- **Languages:** Embedded C, C++, Python, PowerShell
- **Microcontrollers:** ARM Cortex-M4, ARM Cortex-M7
- **RTOS:** embOS, MQX
- **Communication & Interfaces:** CAN, J1939, BLE, UART, RS-232
- **Tools:** J-Link, Kvaser, Peak, BUSMASTER, Git, GCC, IBM Rhapsody, Polarion, IBM Engineering Workflow Management (Jazz/RTC)
- **Practices & Methods:** V-model development, HIL testing, hardware bring-up, requirements traceability, Agile/Scrum
- **Lab Equipment:** Oscilloscope, spectrum analyzer, CAN bus sniffer, multimeter, soldering, harness building